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- لو حابب تشوف تنفيذ المشروع بشكل عملي خطوة بخطوة، تقدر تتابع الفيديو ده:
https://youtu.be/RwFJTJTe-OM?si=UwhZUdikdfy3ochm
الفيديو ده كان مصدر الإلهام الأساسي للمشروع، وأنا قمت بإعادة تنفيذه بنفسي مع ترتيب وتبسيط الشرح في الملف ده بحيث يكون مرجع واضح وسهل للمراجعة.

- كمان وفّرت نسخة من المشروع (Packet Tracer File) على GitHub الخاص بيا تقدر تحملها وتشتغل عليها مباشرة :
https://github.com/NSecurityLab/Networks_Super_Projects

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Project_1: (Hotel Management Network)

As a part of your end year networking project, you are required to design and implement Vic Modern Hotel network. The hotel has three floors; in the first floor there three departments (Reception, store and Logistics), in the second floor there are three departments (Finance, HR and Sales/Marketing), while the third floor hosts the IT and Admin. Therefore, the following are part of the considerations during the design and implementation.

1. There should be three routers connecting each floor (all placed in the server room in IT department).

2. All routers should be connected to each other using serial DCE cable.

3. The network between the routers should be 10.10.10.0/30,10.10.10.4/30,10.10.10.8/30

4. Each floor is expected to have one switch (placed in the respective floor).

5. Each floor is expected to have WIFI networks connected to laptops and phones.

6. Each department is expected to have a printer.

7. Each department is expected to be in different VLAN with the following details:

1st Floor:

- Reception- VLAN 80, Network of 192.168.8.0/24
- Store- VLAN 70, Network of 192.168.7.0/24
- Logistics- VLAN 60, Network of 192.168.6.0/24

2nd Floor:

- Finance- VLAN 50, Network of 192.168.5.0/24
- HR- VLAN 40, Network of 192.168.4.0/24
- Sales- VLAN 30, Network of 192.168.3.0/24

3rd t Floor:

- Admin- VLAN 20, Network of 192.168.2.0/24
- IT- VLAN 10, Network of 192.168.1.0/24

8. Use OSPF as the routing protocol to advertise routes.

9. All devices in the network are expected to obtain IP address dynamically with their respective router configured as the DHCP server.

10. All the devices in the network are expected to communicate with each other.

11. Configure SSH in all the routers for remote login.

12. In IT department, add PC called Test-PC to port fa0/1 and use it to test remote login.

13. Configure port security to IT-dept switch to allow only Test-PC to access port fa0/1 (use sticky method to obtain mac-address with violation mode of shutdown.)

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Solution:

الفكرة العامة للمشروع

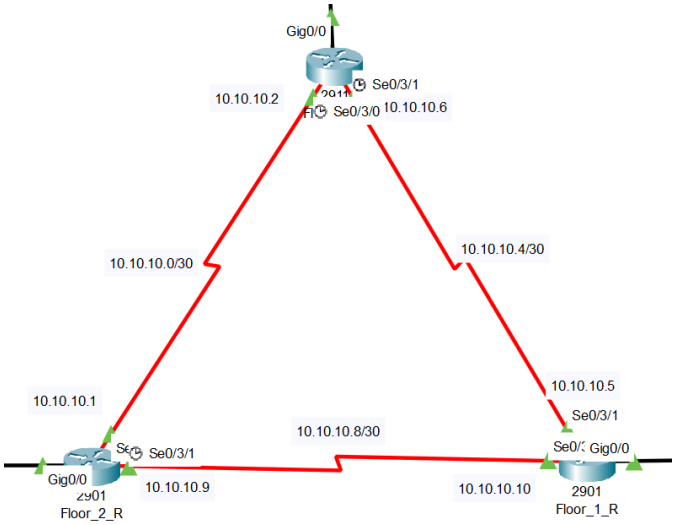
المشروع ده عبارة عن شبكة لفندق مكون من 3 أدوار، وكل دور فيه Departments مختلفة.
إحنا عملنا Design يخلى:

- كل دور له Router خاص بيه
- كل دور فيه Switch
- كل Department ليه VLAN خاصة (عشان السيكيورتى والتنظيم)
- كل الأجهزة تاخد IP أوتوماتيك (DHCP)
- كل الشبكة تكلم بعض (Routing باستخدام OSPF)
- نأمّن الدخول على الراوترات بـ SSH
- نأمّن بورت معين فى (Port Security) IT

Interface Activation on Routers

فى البداية أى Interface فى الراوتر بيكون مقفول(Shutdown) ، فلازم نفتحه عشان يبدأ يشتغل ويعدّى الترافيك بين الأجهزة.

Floor_1_R




```
Router>en
Router#conf t
Router(config)#int s0/3/1
Router(config-if)#no sh
%LINK-5-CHANGED: Interface Serial0/3/1,
changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on
Interface Serial0/3/1, changed state to up

Router(config-if)#int s0/3/0
Router(config-if)#no sh
%LINK-5-CHANGED: Interface Serial0/3/0,
changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on
Interface Serial0/3/0, changed state to up

Router(config-if)#int g0/0
Router(config-if)#no sh
%LINK-5-CHANGED: Interface
GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router#
%SYS-5-CONFIG_I: Configured from console by console
```

Floor_2_R

```
Router>en
Router#conf t
Router(config)#int s0/3/1
Router(config-if)#no sh
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up
```

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```
Router(config-if)#int s0/3/0
Router(config-if)#no sh
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config-if)#int g0/0
Router(config-if)#no sh
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
```

Floor_3_R

```
Router>en
Router#conf t
Router(config)#int s0/3/1
Router(config-if)#no sh
%LINK-5-CHANGED: Interface Serial0/3/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/1, changed state to up

Router(config)#int s0/3/0
Router(config-if)#no sh
%LINK-5-CHANGED: Interface Serial0/3/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/3/0, changed state to up

Router(config)#int gigabitEthernet 0/0
Router(config-if)#no sh
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#ex
Router(config)#do wr
Building configuration...
[OK]
```

Clock Rate Configuration on Serial Interfaces

فى كابل الـ Serial لازم طرف الـ DCE يكون عليه Clock Rate عشان الاتصال يبقى.Stable

---

Floor_2_R

```
Floor_2_R(config)#int s0/3/1
Floor_2_R(config-if)#clock rate 64000
Floor_2_R(config-if)#do wr
```

Floor_3_R

```
Floor_3_R(config)#int s0/3/0
Floor_3_R(config-if)#clock rate 64000
Floor_3_R(config-if)#do wr
```

VLAN Assignment on Switches

الفكرة:

قسمنا كل Department فى VLAN لوحده عشان نقلل الـ Broadcast ونزود الـ.Security

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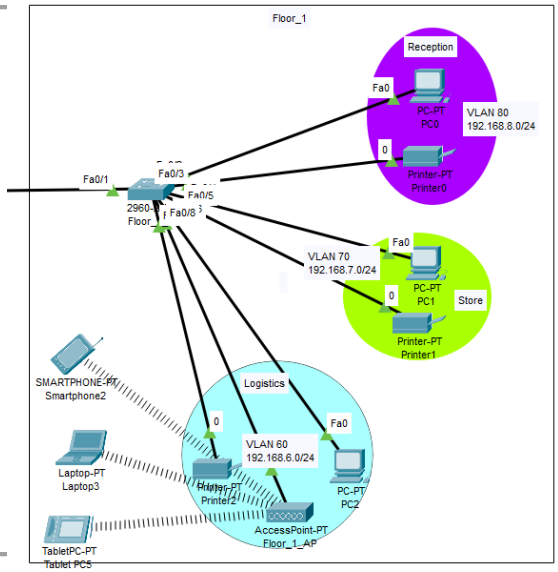
Floor_1_S



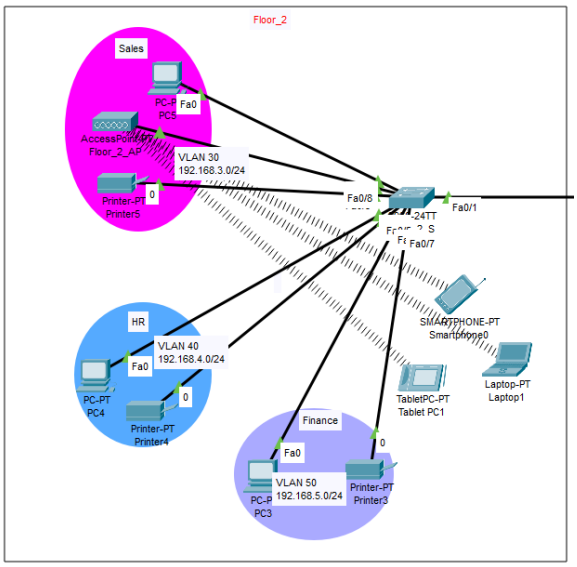

```
Switch(config)#int range f0/2-3
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 80
% Access VLAN does not exist. Creating vlan 80
Switch(config-if-range)#ex

Switch(config)#int range f0/4-5
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 70
% Access VLAN does not exist. Creating vlan 70
Switch(config-if-range)#ex

Switch(config)#int range f0/6-8
Switch(config-if-range)#switchport mode access
Switch(config-if-range)#switchport access vlan 60
% Access VLAN does not exist. Creating vlan 60

Switch(config-if-range)#do wr
```

Floor_2_S




```
Floor_2_S(config)#int r f0/2-3
Floor_2_S(config-if-range)#switchport mode access
Floor_2_S(config-if-range)#switchport access vlan
30
% Access VLAN does not exist. Creating vlan 30

Floor_2_S(config-if-range)#int r f0/4-5
Floor_2_S(config-if-range)#switchport mode access
Floor_2_S(config-if-range)#switchport access vlan
40
% Access VLAN does not exist. Creating vlan 40
Floor_2_S(config-if-range)#ex

Floor_2_S(config)#int r f0/6-7
Floor_2_S(config-if-range)#switchport mode access
Floor_2_S(config-if-range)#switchport access vlan
50
% Access VLAN does not exist. Creating vlan 50

Floor_2_S(config-if-range)#do wr
Building configuration...
[OK]

Floor_2_S(config)#int f0/8
Floor_2_S(config-if)#switchport mode access
Floor_2_S(config-if)#switchport access vlan 30
Floor_2_S(config-if)#ex
Floor_2_S(config)#do wr
```

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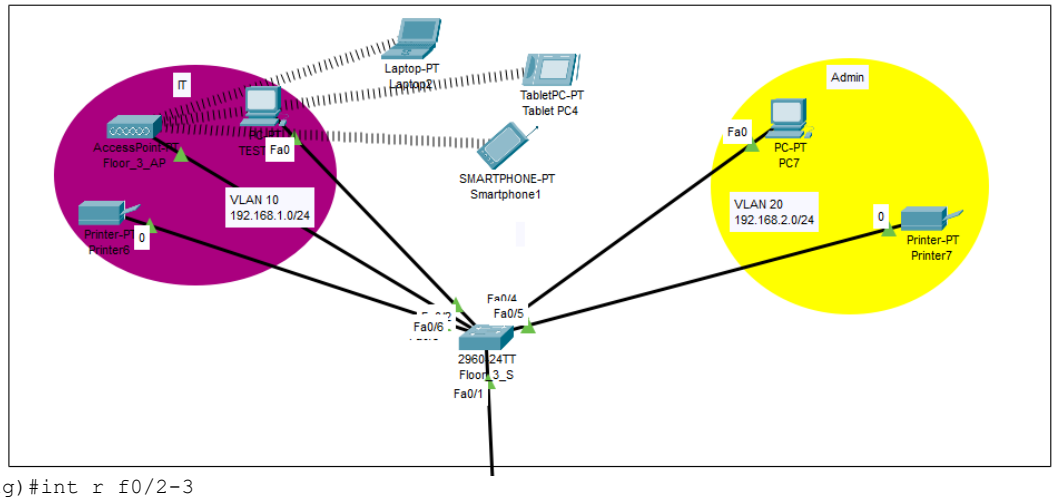
Floor_3_S




```
Floor_3_S(config)#int r f0/2-3
Floor_3_S(config-if-range)#switchport mode access
Floor_3_S(config-if-range)#switchport access vlan 10
% Access VLAN does not exist. Creating vlan 10

Floor_3_S(config-if-range)#int r f0/4-5
Floor_3_S(config-if-range)#switchport mode access
Floor_3_S(config-if-range)#switchport access vlan 20
% Access VLAN does not exist. Creating vlan 20
Floor_3_S(config-if-range)#ex

Floor_3_S(config)#int f0/6
Floor_3_S(config-if)#switchport mode access
Floor_3_S(config-if)#switchport access vlan 10
```

Trunk Mode Configuration on Switch Ports

الفكرة:

بنخلى البورت شايل كل الـ VLANs عشان يوصلها للراوتر.

```
Floor_1_S(config)#int f0/1
Floor_1_S(config-if)#switchport mode trunk

Floor_2_S(config)#int f0/1
Floor_2_S(config-if)#switchport mode trunk

Floor_3_S(config)#int f0/1
Floor_3_S(config-if)#switchport mode trunk
```

---

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IP Address Assignment on Router Interfaces:

الفكرة:

كل راوتر لازم يبقى له IP على اللينكات عشان يتكلم مع باقى الراوترات.

Floor_1_R

```
Floor_1_R#conf t
Floor_1_R(config)#int s0/3/1
Floor_1_R(config-if)#ip address 10.10.10.5 255.255.255.252
Floor_1_R(config-if)#int s0/3/0
Floor_1_R(config-if)#ip address 10.10.10.10 255.255.255.252
Floor_1_R(config-if)#do wr
```

Floor_2_R

```
Floor_2_R(config)#int s0/3/1
Floor_2_R(config-if)#ip address 10.10.10.9 255.255.255.252
Floor_2_R(config)#int s0/3/0
Floor_2_R(config-if)#ip address 10.10.10.1 255.255.255.252
```

Floor_3_R

```
Floor_3_R(config)#int s0/3/0
Floor_3_R(config-if)#ip address 10.10.10.2 255.255.255.252
Floor_3_R(config-if)#int s0/3/1
Floor_3_R(config-if)#ip address 10.10.10.6 255.255.255.252
Floor_3_R(config-if)#do wr
```

Subinterface Creation for VLANs on Routers:

الفكرة:

الراوتر لازم يبقى Default Gateway لكلVLAN ،
بس عندنا بورت واحد → (g0/0) فبنقسمه لعدةSubinterfaces
كل واحدة تمثل VLAN معينة) وده اسمه.(Router-on-a-Stick

Floor_1_R

```
Floor_1_R(config)#int gig0/0.80
Floor_1_R(config-subif)#encapsulation dot1Q 80
Floor_1_R(config-subif)#ip add 192.168.8.1 255.255.255.0
Floor_1_R(config-subif)#ex

Floor_1_R(config)#int gig0/0.70
Floor_1_R(config-subif)#encapsulation dot1Q 70
Floor_1_R(config-subif)#ip add 192.168.7.1 255.255.255.0
Floor_1_R(config-subif)#ex

Floor_1_R(config)#int gig0/0.60
Floor_1_R(config-subif)#encapsulation dot1Q 60
Floor_1_R(config-subif)#ip add 192.168.6.1 255.255.255.0
Floor_1_R(config-subif)#ex
```

---

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DHCP Pool Configuration on Routers

الفكرة:

بدل ما ندى IP لكل جهاز بإيدنا، الراوتر بيوزع IP أوتوماتيك لكل.VLAN

Floor_1_R

```
Floor_1_R(config)#service dhcp

Floor_1_R(config)#ip dhcp pool Reception
Floor_1_R(dhcp-config)#network 192.168.8.0 255.255.255.0
Floor_1_R(dhcp-config)#default-router 192.168.8.1
Floor_1_R(dhcp-config)#dns-server 192.168.8.1
Floor_1_R(dhcp-config)#ex

Floor_1_R(config)#ip dhcp pool Store
Floor_1_R(dhcp-config)#network 192.168.7.0 255.255.255.0
Floor_1_R(dhcp-config)#default-router 192.168.7.1
Floor_1_R(dhcp-config)#dns-server 192.168.7.1
Floor_1_R(dhcp-config)#ex

Floor_1_R(config)#ip dhcp pool Logistics
Floor_1_R(dhcp-config)#network 192.168.6.0 255.255.255.0
Floor_1_R(dhcp-config)#default-router 192.168.6.1
Floor_1_R(dhcp-config)#dns-server 192.168.6.1
Floor_1_R(dhcp-config)#ex

Floor_1_R(config)#do wr
```

OSPF Routing Protocol Configuration

الفكرة:

OSPFده اللى بيخلى كل الراوترات تعرف الشبكات عند بعض
فبالتالى أى جهاز فى أى VLAN يقدر يوصل لأى جهاز تانى.

Floor_1_R

```
Floor_1_R(config)#router ospf 10
Floor_1_R(config-router)#network 10.10.10.8 255.255.255.252 area 0
Floor_1_R(config-router)#network 10.10.10.4 255.255.255.252 area 0
Floor_1_R(config-router)#net 192.168.8.0 255.255.255.0 area 0
Floor_1_R(config-router)#net 192.168.7.0 255.255.255.0 area 0
Floor_1_R(config-router)#net 192.168.6.0 255.255.255.0 area 0
```

Floor_2_R

```
Floor_2_R(config)#router ospf 10
Floor_2_R(config-router)#net 10.10.10.0 255.255.255.252 area 0
Floor_2_R(config-router)#net 10.10.10.8 255.255.255.252 area 0
%OSPF-5-ADJCHG: Process 10, Nbr 192.168.8.1 on Serial0/3/1 from LOADING to FULL, Loading Done
Floor_2_R(config-router)#net 192.168.3.0 255.255.255.0 area 0
Floor_2_R(config-router)#net 192.168.4.0 255.255.255.0 area 0
Floor_2_R(config-router)#net 192.168.5.0 255.255.255.0 area 0
Floor_2_R(config-router)#do wr
```

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Floor_3_R

```
Floor_3_R(config)#router ospf 10
Floor_3_R(config-router)#net 10.10.10.0 255.255.255.252 area 0
%OSPF-5-ADJCHG: Process 10, Nbr 192.168.5.1 on Serial0/3/0 from LOADING to FULL, Loading Done
Floor_3_R(config-router)#net 10.10.10.4 255.255.255.252 area 0
%OSPF-5-ADJCHG: Process 10, Nbr 192.168.8.1 on Serial0/3/1 from LOADING to FULL, Loading Done
Floor_3_R(config-router)#net 192.168.1.0 255.255.255.0 area 0
Floor_3_R(config-router)#net 192.168.2.0 255.255.255.0 area 0
Floor_3_R(config-router)#do wr
```

SSH Configuration on Routers

الفكرة:

بدل ما ندخل على الراوتر بـ Telnet (مش آمن)، بنستخدم SSH لأنه مشفر وآمن.

Floor_1_R

```
Floor_1_R(config)#ip domain-name Someone
Floor_1_R(config)#username Someone password SS123
Floor_1_R(config)#crypto key generate rsa
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Floor_1_R(config)#line vty 0 15
%SSH-5-ENABLED: SSH 1.99 has been enabled
Floor_1_R(config-line)#login local
Floor_1_R(config-line)#transport input ssh
Floor_1_R(config-line)#do wr
```

Floor_2_R

```
Floor_2_R(config)#ip domain-name Someone
Floor_2_R(config)#username Someone password SS123
Floor_2_R(config)#crypto key generate rsa
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Floor_2_R(config)#line vty 0 15
%SSH-5-ENABLED: SSH 1.99 has been enabled
Floor_2_R(config-line)#login local
Floor_2_R(config-line)#transport input ssh
Floor_2_R(config-line)#do wr
```

Floor_3_R

```
Floor_3_R(config)#ip domain-name Someone
Floor_3_R(config)#username Someone password SS123
Floor_3_R(config)#crypto key generate rsa
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

Floor_3_R(config)#line vty 0 15
%SSH-5-ENABLED: SSH 1.99 has been enabled
Floor_3_R(config-line)#login local
Floor_3_R(config-line)#transport input ssh
Floor_3_R(config-line)#do wr
```

---

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Port Security Configuration on IT Switch

الفكرة:

نأمّن بورت معين بحيث يدخل عليه جهاز واحد بس(Test-PC) ، وأى جهاز غريب → البورت يقفل.

Floor_3_S

```
Floor_3_S(config)#int f0/2
Floor_3_S(config-if)#switchport port-security maximum 1
Floor_3_S(config-if)#switchport port-security mac-address sticky
Floor_3_S(config-if)#switchport port-security violation shutdown
Floor_3_S(config-if)#do wr
```